24EEE432 Cyber Physical System for Industrial Applications

Risk-Adaptive & Failure Detection Safety System for Industrial Environments Report

Submitted by

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MARCH 2026

Amrita School of Engineering, Coimbatore

Department of Electrical and Electronics Engineering

Vision

Mould generations of electrical and electronics engineers on global standards with multi-disciplinary perspective to meet evolving societal needs.

Mission

- Empower students with knowledge in Electrical, Electronics and allied Engineering facilitated in innovative classrooms and state-of-the art laboratories.
- Inculcate technical competence and promote research through industry interactions, field exposures and global collaborations.
- Promote professional ethics and selfless service.

Program Educational Objectives (PEOs)

PEO1: Graduate can demonstrate electrical and electronics engineering problem solving skill along with proficiency in communication and professional excellence in project management and execution.

PEO2: Graduate can be employable in engineering services including ICT enabled sectors and also motivated for entrepreneurship.

PEO3: Graduate will be competent for higher studies in world class universities and research in industrial organizations.

PEO4: Graduate will manifest social commitment, environmental awareness and moral and ethical values in professional
and other discourses.

Program Specific Outcome (PSOs)

PSO1: Build and manage electro dynamic systems using Knowledge on electrical technology and semiconductor devices for allied services.
PSO2: Use computational tools and network dynamics for design, analysis and control of power systems integrated with renewable energy and Electric Vehicle.
PSO3: Leverage digital technologies employing state-of-the art control techniques and embedded controllers for industrial applications.

24EEE432 CYBER PHYSICAL SYSTEM FOR INDUSTRIAL APPLICATIONS L-T-P-C: 3-0-0-3

Course Objective:

- Understand and develop hardware and software design requirements of Cyber Physical System for Engineering Applications from Embedded Perspective
- Design and implement a prototype of Embedded Cyber Physical Systems using ARM processor and programming skills in Embedded C, Micro python
- Develop embedded projects for Smart City, Smart Civil Structures, Health Care Application, Automotive Mechatronics, Hybrid Electric Vehicles, Robots for Industrial Automation

Course Outcomes:

CO1: Program for ARM Processor STM32F103 in C language using STM Cube IDE Software Development Environment

CO2: Gain skills for interfacing ARM processor STM32F103 with sensors like Ultrasonic sensor, Hall effect sensor, Tilt Sensor, actuators like servo Motors and displays like OLED and LCD Displays

CO3: Program Raspberry Pi Pico using the Micropython language using Thonny IDE Software Development Environment and develop simple embedded system prototypes using Raspberry Pi Pico to interface with sensors, OLED and LCD displays, motor

CO4: Develop Applications of CPS from embedded perspective in Automotive Mechatronics and Healthcare Applications

CO5: Develop Applications of CPS from embedded perspective in Hybrid Electric Vehicle, Industrial Motor Control and Robotics

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3			3							1	2	3
CO2	3	3										1	2	3
CO3	3	3			3							1	2	3
CO4	3	3	3	2	3		2	2	2	2	1		2	3
CO5	3	3	3	2	3		2	2	2	2	1		2	3

PO1-Engineering Knowledge, PO2-Problem Analysis, PO3-Design/Development of Solutions, PO4-Conduct Investigations of Complex Problems, PO5-Modern Tool Usage, PO6-The Engineer and Society, PO7-Environment and Sustainability, PO8-Ethics, PO9-Individual and Team Work, PO10-Communication, PO11-Project Management and Finance.

Washington Accord Knowledge Profiles (WK) mapping	Strongly Supports POs	Justification
WK2 (Mathematics & Computation)	PO1, PO2	Embedded logic, timing, and control algorithms
WK3 (Engineering Fundamentals)	PO1, PO2, PO3	ARM architecture, sensors, actuators
WK4 (Engineering Specialization)	PO1, PO2, PO4	CPS, Embedded Systems, STM32, Pico
WK5 (Engineering Design)	PO3, PO5	CPS application design & prototyping
WK6 (Engineering Practice)	PO4, PO11	IDEs, debugging, and hardware interfacing
WK7 (Engineering Tools)	PO5	STM Cube IDE, Thonny, MicroPython
WK8 (Ethics, Environment & Sustainability)	PO6, PO7	Healthcare, EV, safety-critical CPS
WK9 (Individual & Team Work)	PO9, PO10	Projects-CPS applications

BONAFIDE CERTIFICATE

This is to certify that the cyber-physical system project report entitled

Risk-Adaptive & Failure Detection Safety System for Industrial Environments

submitted by

Roll Nos.	**Student Names**
CB.EN.U4ECE23004	ADARSH B
CB.EN.U4ECE23016	EZHILKIRTHIK M
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SUBMITTED TO:

Dr. K.R.M. Vijaya Chandrakala

Associate Professor, Department of Electrical and Electronics, School of Engineering, Coimbatore

is in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology** in **"Electrical and Electronics Engineering"** is a Bonafide record of the work carried out at Amrita School of Engineering, Coimbatore.

Internal Examiner External Examiner

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AIM

To design and implement a **Risk-Adaptive & Failure Detection Safety System** for industrial environments that continuously monitors environmental hazards, detects potential risks, controls critical equipment automatically, and ensures secure worker/visitor management using IoT-based authentication and monitoring systems.

PROBLEM STATEMENT

Industrial environments face several critical safety challenges, including fire and smoke hazards, unauthorized access to restricted areas, unsafe proximity to heavy machinery, electrical control room risks, and lack of real-time monitoring and logging.

Conventional safety systems operate independently and lack integrated, adaptive responses. Therefore, there is a need for a smart, automated, and risk-based industrial safety system that can continuously monitor hazards and respond dynamically according to the level of risk.

PROPOSED SOLUTION

We propose a **multi-layer industrial safety system** integrating:

- Fire, smoke, rain, motion, temperature, and humidity detection
- Ultrasonic-based proximity hazard control
- Relay-based machine shutdown
- Servo-based automatic doors
- OTP-based worker & visitor authentication
- Web-based live monitoring and logging
- LCD & OLED real-time display interface

The system automatically adapts safety levels based on risk severity and provides emergency shutdown in danger conditions.

KEY FEATURES OF THE SOLUTION

- **Fire and Smoke Detection with Audible Alert System**

  Continuous monitoring of fire and smoke hazards using dedicated sensors. When

abnormal smoke or heat levels are detected, the system immediately activates a buzzer alarm and visual alert to notify workers and allow early response to prevent industrial accidents.

• **Rain Detection with Automatic Window Control**

A rain sensor continuously monitors environmental conditions. When rain is detected, the system automatically activates servo motors to close industrial windows, preventing water damage to machinery, electrical equipment, and stored materials.

• **Risk-Zone Based Motor Control (40 cm – 10 cm Safety Levels)**

An ultrasonic sensor monitors the distance between workers and hazardous machinery. Multiple safety zones are defined: at 40 cm a warning alert is triggered, at 20 cm machine speed is reduced, and at 10 cm the motor is automatically shut down to prevent injury.

• **Motion-Based Smart Door Automation**

A PIR (Passive Infrared) sensor detects human movement near the entrance. When a person approaches, the door automatically opens using a motor mechanism and closes after a delay, improving convenience while maintaining controlled access.

• **OTP-Based Secure Entry and Exit Authentication**

A One-Time Password (OTP) verification system ensures that only authorized workers and visitors can access the facility. OTPs are generated and validated before granting entry, improving industrial security and preventing unauthorized access.

• **Web-Based Monitoring and Activity Logging Dashboard**

A web-based dashboard allows supervisors to monitor the system in real time through a browser interface. The dashboard displays sensor status, worker entries and exits, safety alerts, and system logs for improved industrial management.

• **Real-Time Worker and Visitor Count Management**

The system automatically maintains a count of workers and visitors present inside the facility. This helps maintain occupancy limits, improves emergency evacuation management, and provides accurate attendance records.

• **Emergency Shutdown and System Lock Mechanism**

Under critical conditions such as fire detection or dangerous proximity to machinery, the system triggers an emergency shutdown procedure. All motors stop immediately and the system enters a safe lock state to prevent further hazards.

• **LCD and OLED-Based Live Status Display**

LCD and OLED display modules provide real-time updates on system status, safety alerts, sensor readings, and operational messages. This allows workers and supervisors to quickly understand the current condition of the industrial environment.

• **Centralized Microcontroller-Based Control System**

All sensors and actuators are integrated through a central microcontroller unit, enabling coordinated monitoring, processing, and control. This ensures reliable operation, quick response to hazards, and efficient system management.

• **Energy Efficient and Automated Operation**

The system operates automatically based on sensor inputs, reducing the need for constant human supervision. Efficient power usage and intelligent control improve reliability and reduce operational costs.

• **Scalable and Modular System Design**

The system architecture is modular, allowing additional sensors, safety modules, or monitoring features to be integrated easily. This makes the solution adaptable for different industrial environments and future upgrades.

PROJECT DESIGN AND ARCHITECTURE

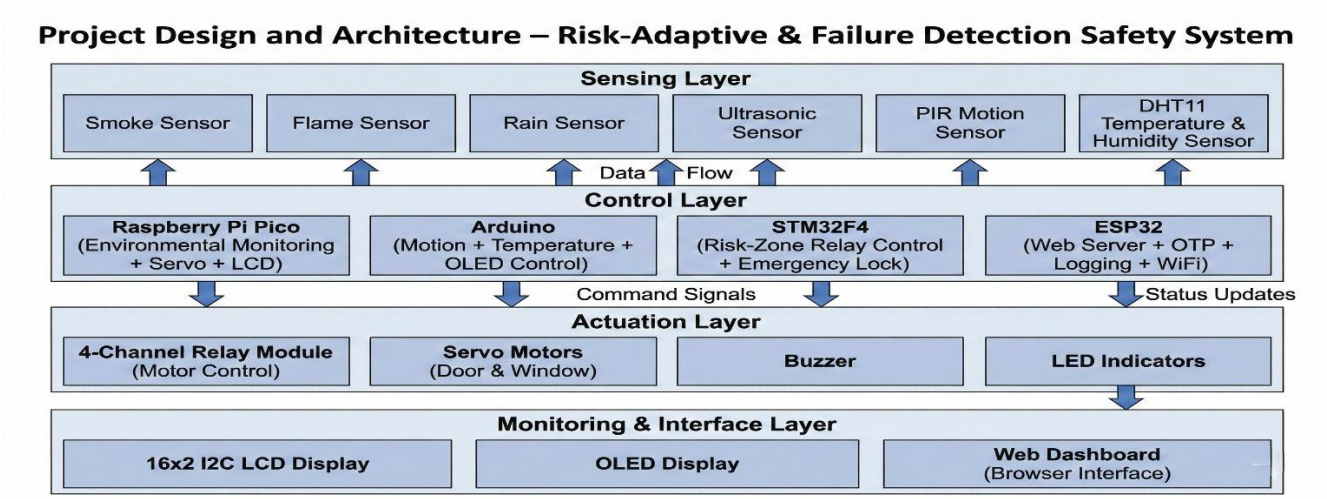


Fig.1 System Architecture

The proposed system follows a **four-layer architecture** to ensure organized operation, modular design, and efficient risk management.

A. Sensing Layer

This layer consists of environmental and safety sensors such as smoke, flame, rain, ultrasonic, PIR motion, and DHT11 temperature-humidity sensors. These sensors continuously monitor industrial conditions and detect potential hazards in real time.

B. Control Layer

The control layer processes sensor data and makes intelligent decisions.

- The **ESP32** handles web communication, OTP authentication, and IoT-based monitoring.
- The **STM32 microcontroller** manages risk-zone-based relay control and emergency shutdown.
- Another microcontroller controls servo motors and LCD display operations.

C. Actuation Layer

Based on control decisions, this layer performs physical actions such as:

- Turning relays ON/OFF for machine control
- Operating servo motors for doors and windows
- Activating buzzer and LEDs for alerts

D. Monitoring Layer

This layer provides real-time system feedback through:

- I2C LCD display
- OLED display
- Web dashboard for live monitoring and activity logging

Overall, the architecture ensures continuous monitoring, intelligent decision-making, automatic hazard response, and secure access control in industrial environments.

SYSTEM OVERVIEW

The **Risk-Adaptive & Failure Detection Safety System for Industrial Environments** is a multi-controller, multi-layer safety platform designed to monitor environmental conditions, control industrial equipment dynamically, and manage secure access within industrial premises.

The system integrates four microcontrollers—Raspberry Pi Pico, Arduino, ESP32, and STM32F4—each assigned specific responsibilities to ensure modularity, reliability, and efficient distributed processing.

Environmental Monitoring and Hazard Detection

The system continuously monitors critical environmental parameters such as smoke, flame, rain, temperature, humidity, and motion. The Raspberry Pi Pico and Arduino collect real-time data from respective sensors and trigger immediate responses when abnormal conditions are detected. In case of fire or smoke detection, audible alerts are activated and emergency exits are automatically opened. Rain detection initiates automatic window control to prevent environmental damage.

Risk-Adaptive Motor Control

The STM32F4 microcontroller manages industrial equipment safety using ultrasonic-based proximity sensing. The monitored distance is categorized into predefined safety zones (40 cm, 30 cm, 20 cm, and 10 cm). As a person approaches hazardous machinery, relays are gradually deactivated to reduce operational risk. If the critical danger threshold is reached, the system performs a complete emergency shutdown and activates a safety lock mechanism to prevent accidents.

Smart Access Control and Monitoring

The ESP32 module handles secure access management using OTP-based authentication for workers and visitors. It generates and verifies one-time passwords, assigns temporary visitor IDs, and maintains real-time logs of all entries and exits. A web-based dashboard enables live monitoring of system status, occupancy count, and activity logs.

Real-Time Display and User Interface

The system provides on-site visual feedback through an I2C LCD and OLED display. These interfaces display safety status, alerts, environmental data, and authentication information, ensuring clear communication of system conditions to users.

Overall, the system combines environmental safety monitoring, adaptive equipment control, and secure access management into a unified industrial safety framework, enhancing operational reliability and preventing accidents in high-risk industrial environments.

WORKING PRINCIPLE

The Risk-Adaptive & Failure Detection Safety System operates on a multi-controller distributed processing model. The system's functionality is divided into four distinct operational phases that transition from environmental sensing to intelligent actuation and remote monitoring.

1. Environmental Hazard Sensing and Adaptive Response

The system maintains a continuous monitoring loop through the Raspberry Pi Pico and Arduino modules.

- **Fire & Smoke:** The MQ-series smoke sensor and IR flame sensor provide analog inputs to the Pico; upon detection, the system triggers a buzzer alert and utilizes a servo motor to automatically open emergency exits.
- **Climate Control:** The rain sensor detects moisture levels, prompting the system to initiate automatic window closure via servo actuation to protect internal industrial equipment.
- **Ambient Monitoring:** DHT11 sensors track temperature and humidity, which are processed by the Arduino and updated in real-time on the 0.96" OLED display for site supervisors.

2. Risk-Adaptive Proximity and Machine Safety

The STM32F4 microcontroller manages high-risk machinery safety through an ultrasonic-based distance calculation engine.

- **Zone Categorization:** The system defines four critical safety zones: 40 cm (Safe), 30 cm (Zone 1), 20 cm (Zone 2), and 10 cm (Danger).
- **Progressive Deactivation:** As the HC-SR04 sensor detects human proximity, the STM32F4 sends signals to a 4-channel relay module to gradually shut down specific machinery components to mitigate risk.
- **Critical Shutdown:** If the 10 cm threshold is breached, the system executes an immediate full-system lock and emergency shutdown to prevent mechanical accidents.

3. Secure Access Control and Identity Management

Authorized entry is governed by the ESP32 module, which acts as a centralized gatekeeper.

- **Two-Factor Authentication:** Workers provide a predefined ID, which triggers the generation of a unique One-Time Password (OTP).
- **Visitor Protocol:** Temporary visitor IDs are allocated and validated through the same OTP verification process to prevent unauthorized access to restricted zones.
- **PIR Automation:** Once authenticated, PIR motion sensors detect physical presence to open the smart door servos for a timed interval.

4. IoT Integration and Data Logging

The communication layer leverages the ESP32's built-in WiFi to bridge hardware actions with digital supervision.

- **Web Dashboard:** All entry/exit events, occupancy counts, and sensor alerts are transmitted to a web-based browser interface for remote monitoring.
- **Real-Time Visualization:** Critical system statuses are simultaneously mirrored on a 16x2 I2C LCD and an OLED display, ensuring that on-site personnel have immediate visibility of environmental and operational safety levels.

CIRCUIT DIAGRAM

ESP32:

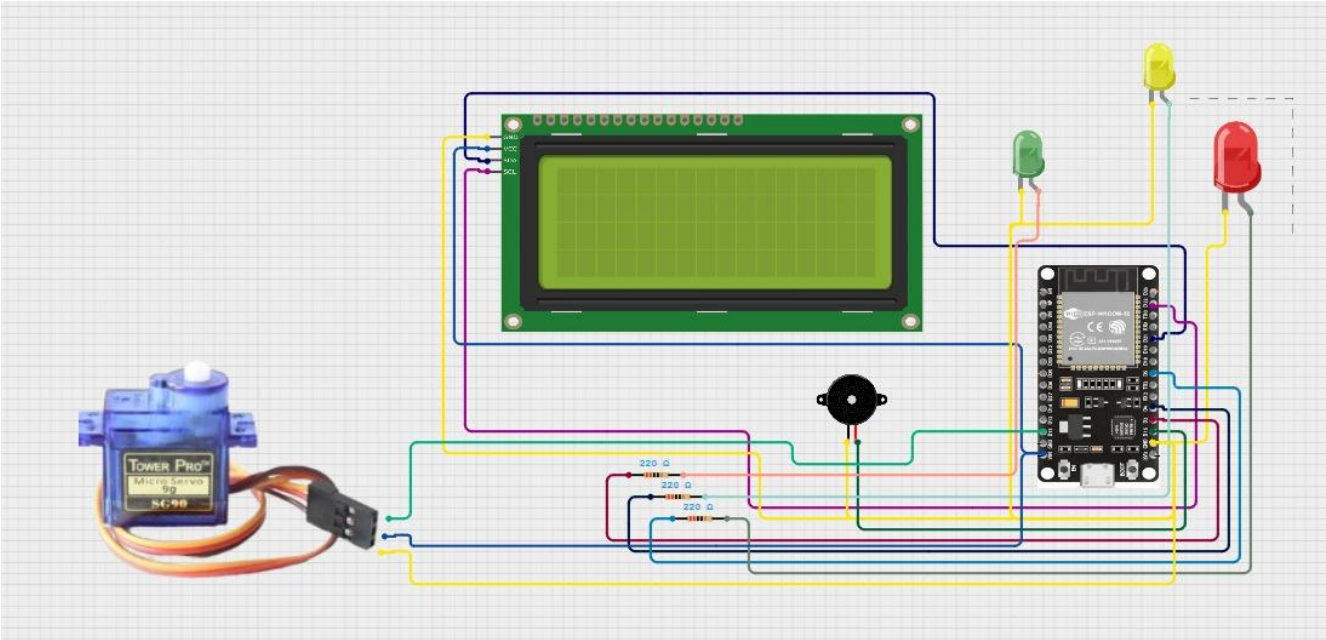


Fig.2 ESP32 Schematic

STM32:

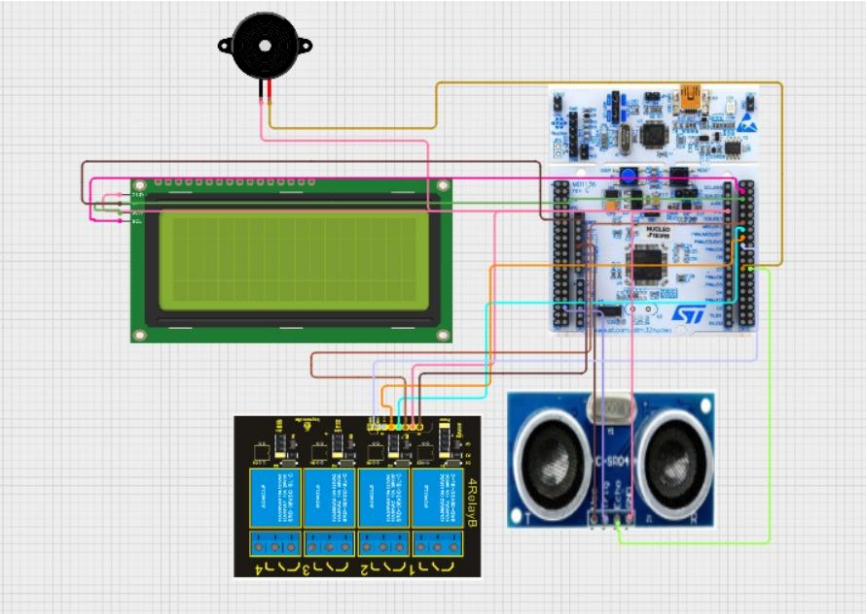


Fig.3 STM32 Schematic

RASPBERRY PI PICO:

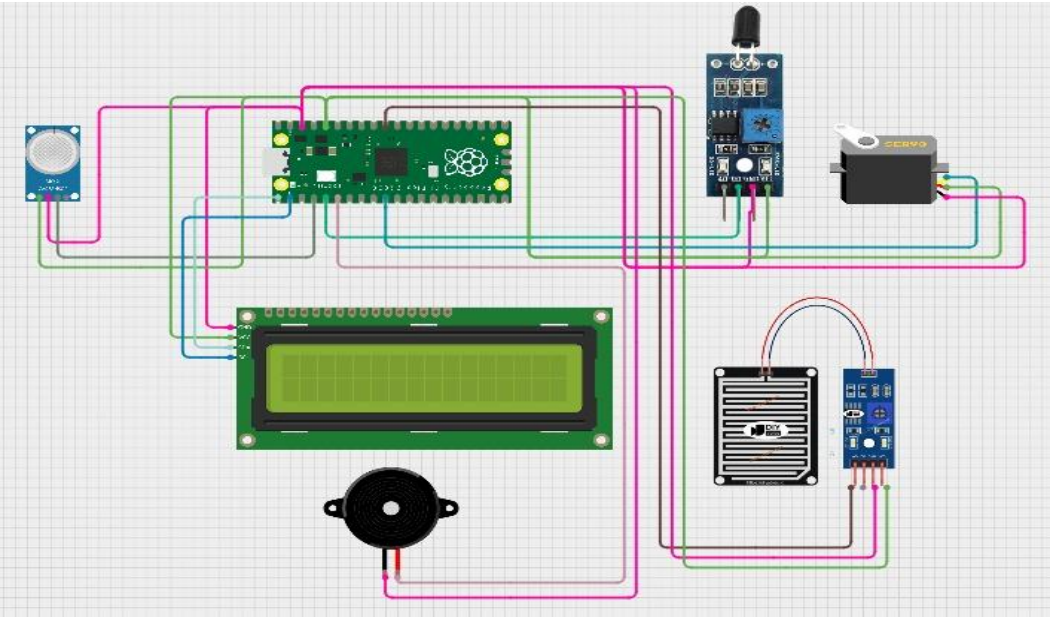


Fig.4 Raspberry Pi Pico Schematic

ARDUINO :

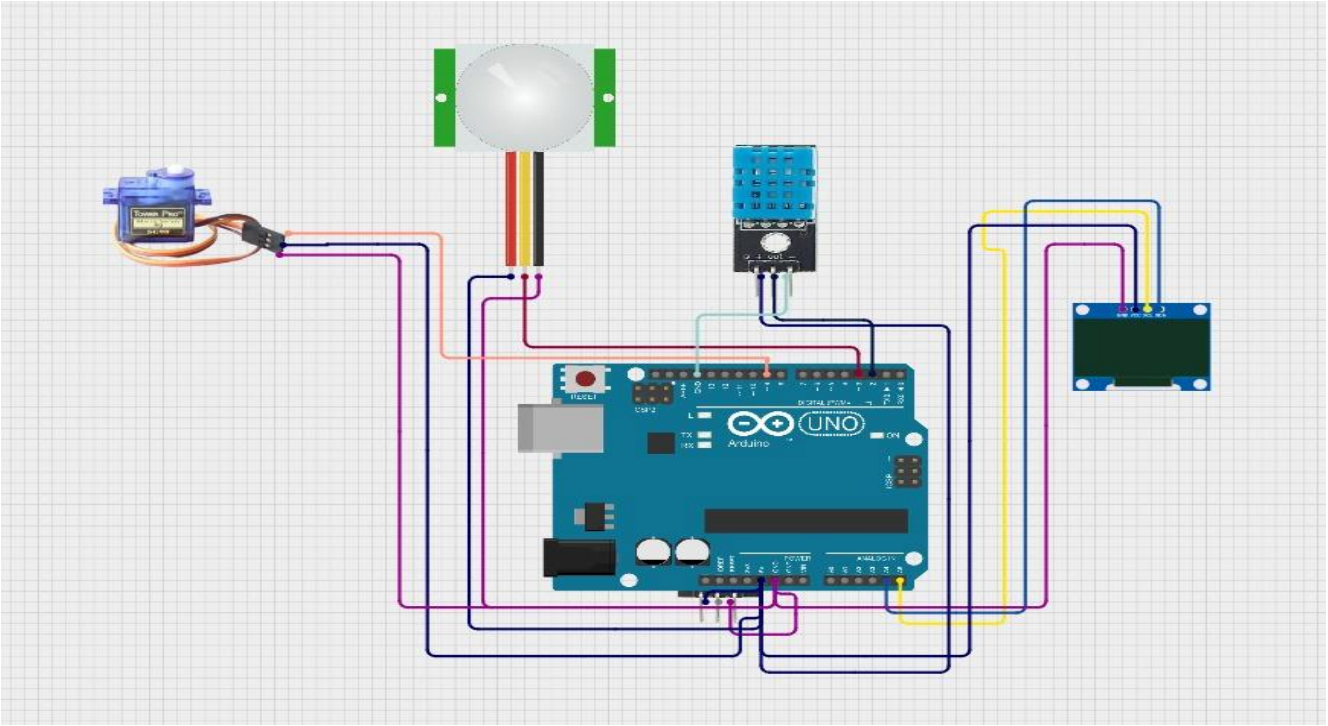


Fig.5 ARDUINO Schematic

PINOUT:

Component	Pin Name	ESP32 GPIO
LCD	SDA	21
LCD	SCL	22
Servo	Signal	13
Buzzer	+	15
Green LED	Anode	2
Yellow LED	Anode	4
Red LED	Anode	5

Table 1. ESP32 Pinout

Component	Pin Type	STM32 Pin
LCD SDA	I2C1 SDA	PB9
LCD SCL	I2C1 SCL	PB8
Ultrasonic TRIG	Digital Out	PB0
Ultrasonic ECHO	Digital In	PB1
Relay 1	Digital Out	PA5
Relay 2	Digital Out	PA6
Relay 3	Digital Out	PA7
Relay 4	Digital Out	PB6
Buzzer	Digital Out	PA8
Reset Button	Digital In	PC13

Table 2. STM32 Pinout

Component	Pin Type	Pico GPIO
LCD SDA	I2C	GP0
LCD SCL	I2C	GP1
Smoke DO	Digital In	GP2
Flame DO	Digital In	GP3
Buzzer	PWM Out	GP4
Exit Servo	PWM Out	GP6
Window Servo	PWM Out	GP7
Rain AO	ADC In	GP26

Table 3. Raspberry Pi Pico Pinout

Component	Pin Name	Arduino Pin
OLED SSD1306	SDA	A4
OLED SSD1306	SCL	A5
DHT11	DATA	D2
PIR HC-SR501	OUT	D3
Servo Motor	Signal	D9

Table 4. Arduino Pinout

HARDWARE

Microcontrollers

- **Raspberry Pi Pico (RP2040)** – Environmental monitoring and servo control
- **Arduino (UNO/Nano)** – Motion detection, temperature/humidity monitoring, OLED control
- **ESP32** – Web server, OTP authentication, WiFi communication
- **STM32F4 Microcontroller** – Risk-zone relay control and emergency shutdown

Sensors

- **Smoke Sensor (MQ Series)** – Detects smoke and gas leakage
- **Flame Sensor** – Detects fire presence
- **Rain Sensor Module** – Detects rainfall/moisture
- **Ultrasonic Sensor (HC-SR04)** – Measures distance for risk-zone detection
- **PIR Motion Sensor** – Detects human movement
- **DHT11 Sensor** – Measures temperature and humidity

Actuators

- **4-Channel Relay Module** – Controls industrial motors and equipment
- **Servo Motors (2–3 units)** –
    - Emergency exit door
    - Smart door
    - Window control
- **Buzzer** – Audible warning system
- **LED Indicators (Red, Yellow, Green)** – Status indication

Display Units

- **16x2 I2C LCD Display** – System status and alerts
- **0.96' OLED Display (SSD1306)** – Environmental data and motion display

Communication & Networking

- **WiFi (ESP32 Built-in Module)** – Web dashboard and remote monitoring

Power & Supporting Components

- 5V Power Supply
- 3.3V Power Regulation
- Breadboard / PCB
- Jumper Wires (Male-Male, Male-Female)
- Resistors
- Push Button (System Reset – STM32 module)

SOFTWARE ARCHITECTURE

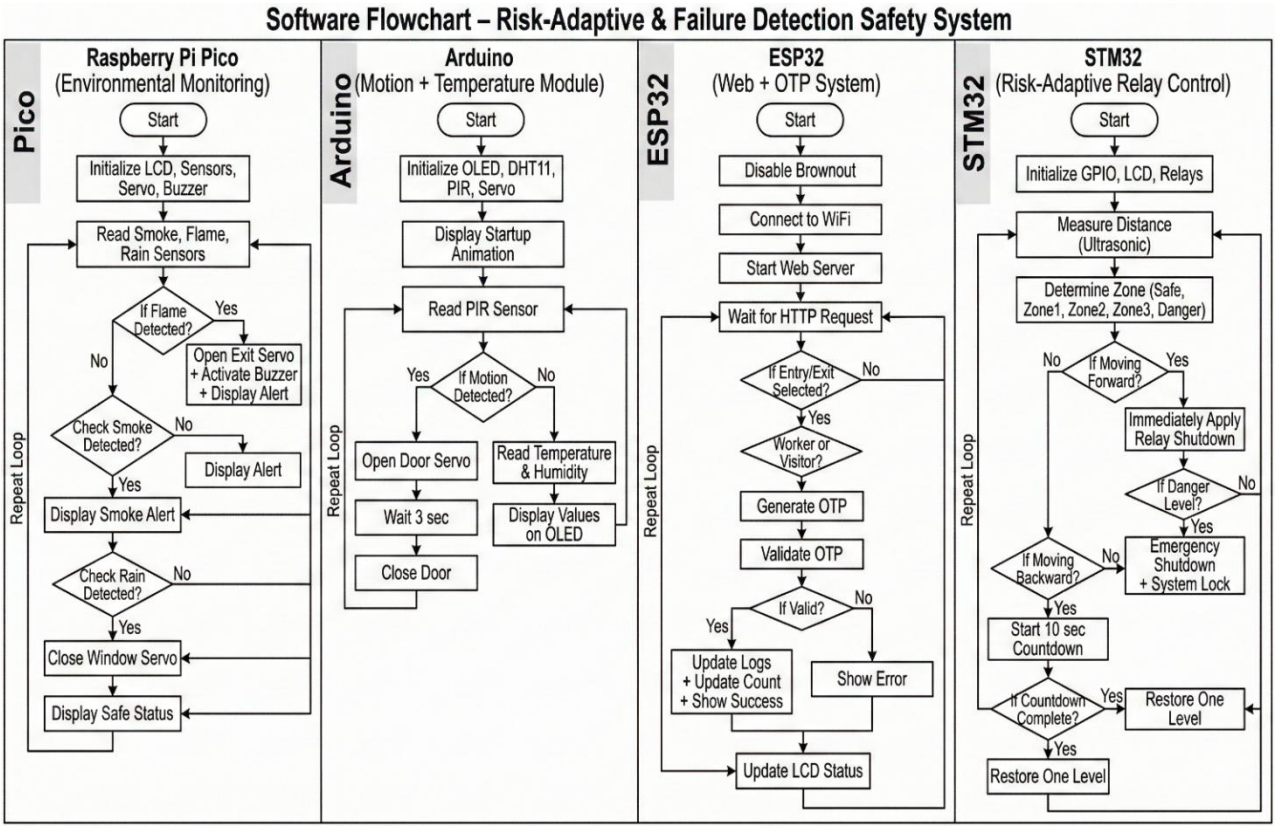


Fig.6 Block Diagram

Programming Languages Used:

- Embedded C (STM32 HAL)
- Arduino C++ (Servo + OLED + DHT)
- MicroPython (ESP32 Web Server+ Raspberry pi pico)

Functional Modules:

- Sensor Interface Module
- Relay Control Module
- Risk-Level Engine
- OTP Generator Module
- Web Server & HTTP Handler
- Logging System
- LCD Display

BILL OF MATERIALS (BOM)

S.No	Component	Specification	Qty	Approx. Cost (₹)	Total (₹)
1	Raspberry Pi Pico	RP2040	1	350	350
2	ESP32 Dev Board	WiFi + Bluetooth	1	500	500
3	STM32F4 Board	ARM Cortex-M4	1	900	900
4	Arduino UNO/Nano	ATmega328P	1	600	600
5	Ultrasonic Sensor	HC-SR04	1	120	120
6	Smoke Sensor	MQ Series	1	180	180
7	Flame Sensor	IR Module	1	120	120
8	Rain Sensor Module	Analog/Digital	1	120	120
9	PIR Motion Sensor	HC-SR501	1	150	150
10	DHT11 Sensor	Temp & Humidity	1	120	120
11	4-Channel Relay Module	5V	1	250	250
12	Servo Motors	SG90	4	150	600
13	Buzzer	5V Active	1	10	10
14	LED Indicators	Red/Yellow/Green	3	2	6
15	16x2 LCD (I2C)	Display	3	250	750
16	OLED Display	0.96’ SSD1306	1	250	250
17	Push Button	STM32 INBUILT	1	-	-
18	Power Supply	5V Adapter	1	200	200
19	Voltage Regulator	3.3V	1	50	50
20	Breadboard / PCB	Prototyping	4	50	200
21	Jumper Wires	Set	100	1.5	150
22	Resistors & Misc.	Passive components	-	100	100
				TOTAL	5,726

RESULTS AND DISCUSSION

Results

The proposed Risk-Adaptive & Failure Detection Safety System was successfully designed, implemented, and tested under various simulated industrial conditions. The system demonstrated reliable performance across all functional modules.

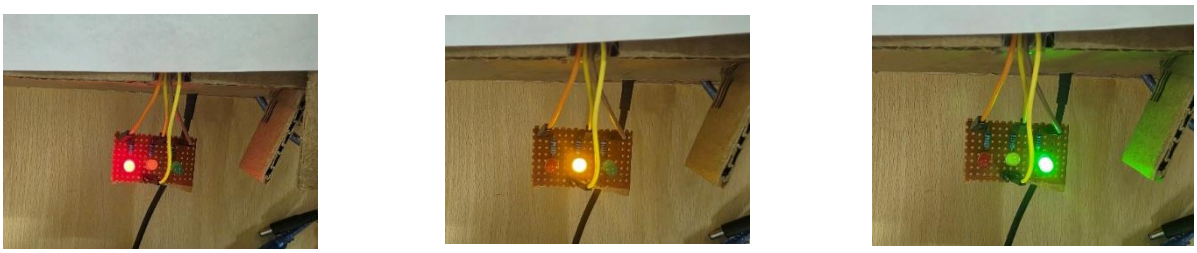
The Environmental Safety Module accurately detected fire, smoke, and rain conditions. Upon flame detection, the system immediately activated the buzzer and opened the emergency exit mechanism. Smoke detection triggered warning alerts, while rain detection successfully controlled automatic window closure.

The Risk-Adaptive Motor Control Module effectively categorized proximity into predefined safety zones (40 cm, 30 cm, 20 cm, and 10 cm). Relay deactivation occurred progressively as the object approached the restricted zone. At the critical danger level (below 10 cm), the system executed a complete emergency shutdown and activated the safety lock mechanism, ensuring maximum protection.

The Smart Access Control Module successfully authenticated workers and visitors using OTP-based verification. The system generated secure one-time passwords, maintained real-time entry and exit logs, and accurately tracked occupancy counts through the web dashboard.

The LCD and OLED displays provided continuous real-time system updates, improving on-site visibility and operational awareness.

ESP32 :



(a) (b) (c)

Fig.7 LED Indication for Entry/Exit

(a)

(b)

(c)

Fig.8 LCD Display Unit

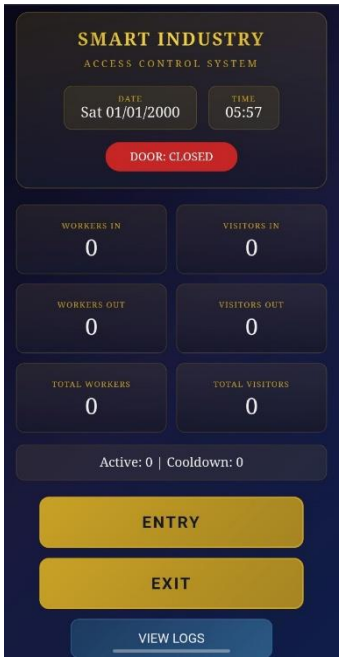
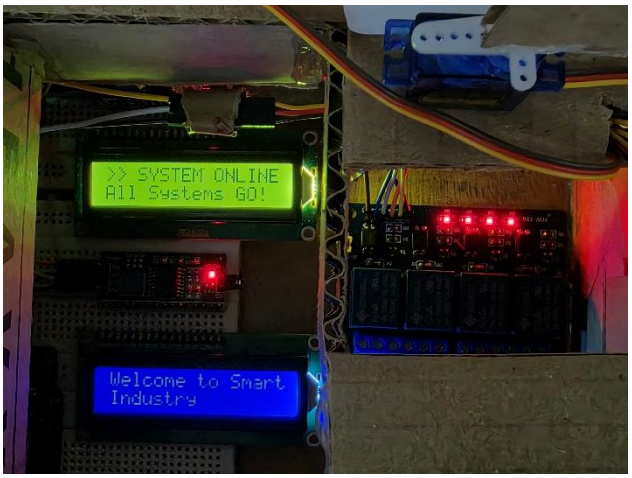


Fig.9 Web Interface

STM32:

(a)



(b)

(c)

Fig.10 Proximity alert with LCD Display Unit

RASPBERRY PI PICO:

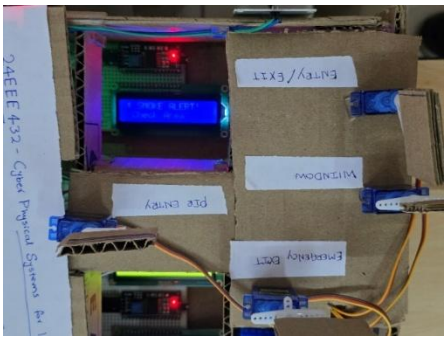


Fig.11 Motor Unit and LCD Display Unit

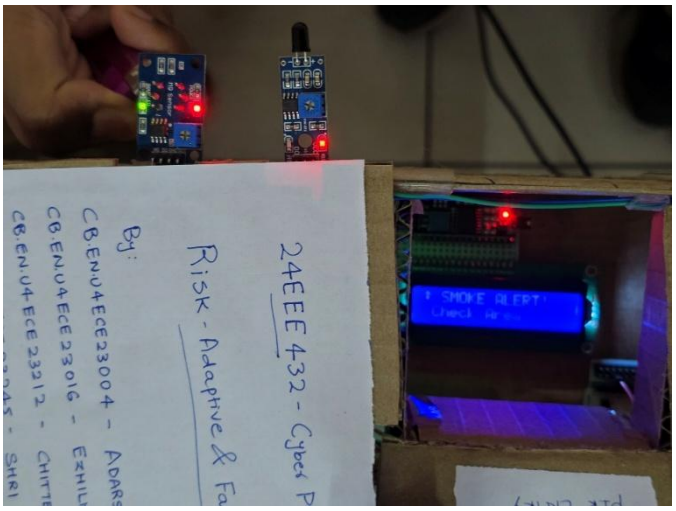
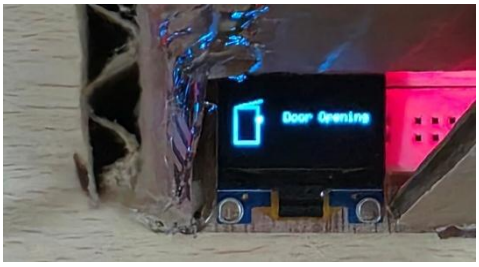


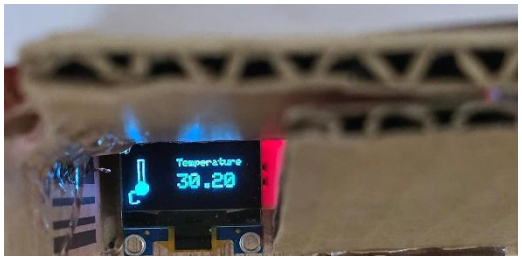
Fig.12 Sensor Unit

ARDUINO:

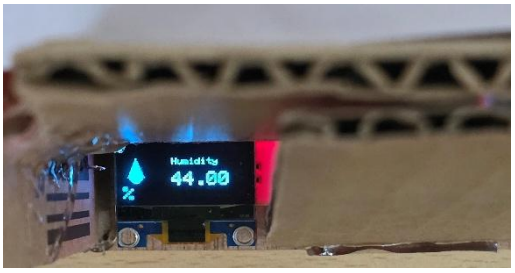


(a)

(b)



(c)



(d)

Fig.13 PIR and DHT 11 with OLED Display Unit

Discussion

The experimental results confirm that the system effectively integrates environmental monitoring, adaptive risk management, and secure access control into a unified safety platform. The distributed multi-controller architecture enhanced system stability and ensured that failure in one module did not disrupt the overall operation.

The risk-zone-based adaptive shutdown strategy proved to be particularly effective in preventing unsafe proximity to machinery. The 10-second controlled backward delay mechanism ensured smooth recovery without sudden system restarts.

The OTP-based authentication system improved industrial security by preventing unauthorized access and maintaining proper activity logs. The web-based monitoring interface enhanced transparency and allowed real-time supervision.

Overall, the system demonstrated high reliability, fast response time, modular scalability, and practical applicability in industrial environments. The results validate the effectiveness of integrating embedded systems and IoT technologies to enhance industrial safety and operational efficiency.

FUTURE SCOPE AND APPLICATIONS

Future Scope

Although the proposed system successfully integrates environmental monitoring, adaptive risk control, and secure access management, several enhancements can further improve its functionality and scalability:

- **Integration with Cloud Platforms:** Storing logs and sensor data on cloud servers for long-term analysis and remote supervision.
- **AI-Based Risk Prediction:** Implementing machine learning algorithms to predict hazardous conditions before they occur.
- **Face Recognition Authentication:** Replacing OTP-based access control with biometric or facial recognition systems for higher security.
- **Mobile Application Integration:** Developing a mobile app for real-time notifications, alerts, and monitoring.
- **GSM/SMS Alert System:** Sending emergency alerts directly to supervisors during critical conditions.
- **Gas Leakage Detection Integration:** Adding industrial gas sensors for enhanced safety in chemical environments.
- **Predictive Maintenance Analytics:** Monitoring equipment usage data to prevent mechanical failures in advance.

These improvements can transform the system into a fully intelligent, Industry 4.0-compliant safety solution.

Applications

The proposed system can be effectively deployed in various industrial and high-risk environments, including:

- Motor Control Rooms
- Manufacturing Industries
- Power Plants
- Chemical Processing Units
- Warehouses and Storage Facilities
- Smart Factories
- Research Laboratories
- Construction Sites

The system enhances industrial safety, prevents accidents, ensures secure access control, and improves operational efficiency across diverse applications.

PROJECT OUTCOMES

The implementation of the Risk-Adaptive & Failure Detection Safety System achieves the following technical and operational results:

- **Integrated Multi-Hazard Monitoring**: Successfully establishes a continuous sensing loop that detects fire, smoke, and rain, triggering immediate protective actuations such as emergency exit release and automatic window closure.
- **Adaptive Machine Risk Mitigation**: Realizes a dynamic proximity-based safety framework using ultrasonic sensing to categorize distances into four safety zones (40 cm to 10 cm), enabling progressive machinery deactivation and total emergency shutdown during critical breaches.
- **Secure Access and Accountability**: Provides a robust worker and visitor management system through OTP-based authentication, ensuring that only verified personnel enter restricted industrial zones while maintaining real-time occupancy counts.

- **Distributed Processing Reliability**: Utilizes a four-microcontroller architecture (Pico, Arduino, ESP32, STM32F4) to ensure modularity, where the failure of a single monitoring component does not compromise the core emergency shutdown functions.
- **Enhanced Operational Transparency**: Delivers real-time situational awareness through a dual-display interface (LCD and OLED) and a centralized web dashboard for remote activity logging and system status supervision.
- **System Stability and Recovery**: Incorporates a 10-second controlled backward delay mechanism to prevent sudden electrical surges or mechanical stress during system restarts after a safety intervention.

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CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

Title of the Project: Risk-Adaptive & Failure Detection Safety System for Industrial Environments

Project Member(s) Register Number: CB.EN.U4ECE23004, CB.EN.U4ECE23016, CB.EN.U4ECE23212, CB.EN.U4ECE23245

Guide Name & Designation: Dr. K.R.M. Vijaya Chandrakala

SDGs	Justification to the Contribution
9: Industry, Innovation and Infrastructure	Deploys a modular IoT architecture (STM32/ESP32) to modernize industrial infrastructure with automated hazard response
3: Good Health and Well-being	Prevents occupational injuries through ultrasonic proximity sensing and real-time monitoring of fire, smoke, and gas.
8: Decent Work and Economic Growth	Secures the workspace via authentication and protects economic assets by preventing machinery accidents.
11: Sustainable Cities and Communities	The integration of smart factory technologies contributes to safer industrial zones, which are critical components of sustainable urban infrastructure.
12: Responsible Consumption and Production	Reduces resource waste & equipment loss through failure detection and emergency shutdown engine.